

Reconfigurable Intelligent Surface Aided Mixed Source Localization by a Single Receiving Antenna

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Abstract—A symmetric reconfigurable intelligent surface (RIS) assisted algorithm for mixed near-field (NF) and far-field (FF) source localization utilizing a single receiving antenna is proposed, focusing on the reconstruction of the spatial covariance matrix (SCM). In detail, by exploiting the geometric symmetry of the RIS and a carefully designed time-varying reflection phase pattern, the SCM is first reconstructed through linear operations. Leveraging the distinct structural properties of FF and NF covariance matrices, a spatial differencing technique is employed to effectively classify mixed source types, and subsequently achieve NF source localization via spectral. Furthermore, an oblique projection method is incorporated to suppress NF interference and enable reliable FF direction-of-arrival (DOA) estimation. Compared with existing approaches, the proposed method achieves significantly improved signal classification accuracy and localization performance while maintaining low hardware complexity. Numerical simulations are provided to validate the effectiveness of the proposed algorithm.

Index Terms—Mixed source localization, reconfigurable intelligent surface (RIS), spatial covariance matrix (SCM) reconstruction, single receiving antenna.

I. INTRODUCTION

In wireless communication and radar detection systems, mixed near-field (NF) and far-field (FF) source localization is a fundamental requirement for accurate target sensing and situational awareness [1] [2]. So far, several methods under different frameworks have been investigated, and representative approaches include the two-stage multiple signal classification (TSMUSIC) method [3], the covariance-based differencing and projection method [4], the sparse signal reconstruction (SSR) and compressed sensing (CS) based method [5]. However, all these methods are established for scenarios of line-of-sight (LOS) propagation and rely on multiple antennas.

However, in wireless communications, non-line-of-sight (NLOS) transmission is a common occurrence. In such circumstances, existing LOS transmission-based mixed source localization methods become ineffective due to model mismatch problems. In recent years, reconfigurable intelligent surfaces (RIS) have been widely regarded as a potential solution, as they can help establish new or virtual LOS links in the presence of obstructions and provide innovative solutions for complex localization challenges by leveraging their electromagnetic environment modulation capabilities [6].

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Motivated by the advantages of RIS, several solutions for enhancing wireless communications and source localization have been proposed recently. In [7], a multi-user channel estimation and localization scheme based on a circular RIS architecture is introduced, achieving parameter decoupling through tensor decomposition and subspace estimation. In [8], a near-field RIS-assisted localization model was established, incorporating joint localization and mutual coupling parameter estimation. These methods represent significant attempts at source localization in NLOS scenarios. However, it is important to note that these approaches are based solely on either pure FF or pure NF source scenarios, rendering them unsuitable for complex mixed-field environments. Additionally, these methods require the receiving antenna array to be configured at the receiver. As a result, they struggle to balance localization accuracy with cost-effectiveness, making them less suitable for practical applications.

In this paper, a cost-effective and high-precision mixed NF and FF source localization scheme for a single receiving antenna is proposed, with the assistance of a symmetric RIS. The proposed method first reconstructs a spatial covariance matrix (SCM) by leveraging the adjustable phase characteristics of the symmetric RIS. It then employs a spatial differencing technique to classify signal types, subsequently achieving NF source localization through spectral search. Finally, by applying the oblique projection technique to extract the NF covariance matrix, the direction of arrival (DOA) estimation for the FF sources is obtained. The proposed method reduces the hardware costs of the system and significantly improves source localization performance, providing both theoretical significance and practical application value. Numerical simulation results validate the effectiveness of the proposed method.

II. PROBLEM FORMULATION

Suppose that K narrowband and uncorrelated source signals (including K_1 NF and K_2 FF source signals) impinge on a symmetric RIS array consisting of $L = 2M + 1$ elements, and a single receiving antenna is employed to collect the reflected signals as depicted in Fig. 1. Without loss of generality, the array center is chosen as the phase reference point, and the distance between adjacent reflective surface elements is d . By assumption, the RIS array center is located at the origin [9], yielding that the position of the l -th RIS unit (where $l =$

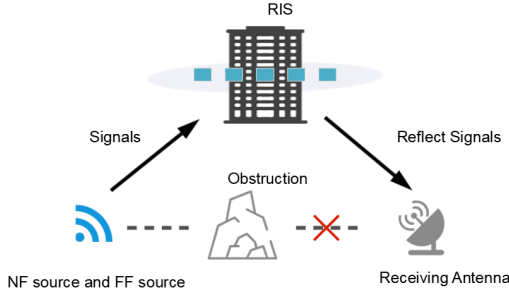


Fig. 1. architecture of RIS-assisted mixed-source localization.

$-M, \dots, 0, \dots, M$ is ld . Consequently, the received signals on the l -th RIS unit can be expressed as

$$x_l(t) = \sum_{k=1}^K s_k(t) e^{j\tau_{lk}} + n_l(t) \quad (1)$$

where $s_k(t)$ represents a transmitted signal emitted from the k -th target and $n_l(t)$ is additive Gaussian white noise with variance σ^2 , τ_{lk} represents the delay related to the k -th signal propagation time from the 0-th RIS unit to the l -th RIS unit, which is given by

$$\tau_{lk} = l\gamma_k + l^2\phi_k \quad (2)$$

where γ_k and ϕ_k are called electric angles, given by

$$\gamma_k = -2\pi \frac{d}{\lambda} \sin \theta_k \quad (3)$$

$$\phi_k = \pi \frac{d^2}{\lambda r_k} \cos^2 \theta_k \quad (4)$$

with λ representing the carrier wavelength of the source signal. In general, if the k -th source is a FF one, τ_{lk} will reduce to $l\gamma_k$; otherwise, if it is a NF one, θ_k and r_k stand for its DOA and range, respectively. According to the definition of NF source, r_k should be within the Fresnel region belonging to $[0.62(D^2/\lambda)^{1/2}, 2D^2/\lambda]$, where $D = (L-1)d$ is the physical array aperture [11].

Collecting the received signals of entire RIS array to yield a matrix form, we have

$$\mathbf{x}(t) = \mathbf{A}_N \mathbf{s}_N(t) + \mathbf{A}_F \mathbf{s}_F(t) + \mathbf{n}(t) \quad (5)$$

where $\mathbf{A}_N$ and $\mathbf{A}_F$ represent the array manifold matrices with respect to NF and FF sources, respectively, $\mathbf{s}_N(t)$ and $\mathbf{s}_F(t)$ are their corresponding source signals.

After reflection by RIS, the signal received by the receiver at the p -th measurement with the g -th candidate RIS shift angle can be expressed as

$$y_{g,p}(t) = \mathbf{h}^T \Phi (\mathbf{A}_N \mathbf{s}_N(t) + \mathbf{A}_F \mathbf{s}_F(t)) + \bar{\mathbf{n}}(t) \quad (6)$$

where $\mathbf{h}$ denote the channel vector from the RIS to the receiver, $\bar{\mathbf{n}}(t) = \mathbf{h}^T \mathbf{n}(t)$ and Φ represent the RIS control matrix, given by

$$\Phi = \text{diag}(\mathbf{u}_g) = \text{diag}([A_{1,g} e^{j\varphi_{1,g}}, \dots, A_{L,g} e^{j\varphi_{L,g}}]) \quad (7)$$

with $\mathbf{u}_g$ standing for the RIS amplitude-phase shifting vector.

Suppose that $\mathbf{h}$ is known *a priori*, by introducing an adjustment vector $\mathbf{b}$ that satisfies $\Phi = \text{diag}(\mathbf{b} \odot \mathbf{u}_g)$ with $\mathbf{b} \odot \mathbf{h}_r^* = \mathbf{1}$ [10]. Meanwhile, $A_{m,g}$ is set to 1 to maximize the signal reflection. Subsequently, after collecting P measurements to yield an output vector at the receiver side, we have

$$\mathbf{y}_g = [y_{g,1}, y_{g,2}, \dots, y_{g,P}]^T = \sum_{l=1}^L \sum_{k=1}^K s_k e^{j\tau_{lk}} \begin{bmatrix} e^{j\varphi_{l,g}} \\ \vdots \\ e^{j\varphi_{l,g}} \end{bmatrix} + \bar{\mathbf{n}} \quad (8)$$

III. SCM RECONSTRUCTION

The single-antenna receiving scheme is confronted with fundamental challenges: it loses the capability of performing spatial spectrum estimation via traditional array processing. The hybrid signals observed by a single antenna are mixed in the time-frequency domain, and the conventional covariance matrix degenerates into a scalar due to the single receiving dimension, failing to carry sufficient angular resolution information. To address these issues, this paper proposes a SCM reconstruction based method that effectively overcomes the aforementioned challenges and significantly improves the localization estimation performance.

To reconstruct the overall SCM, we need to use P available measurements with phase shift φ_g and calculate the average power of the output of a single receiving antenna by the following formula

$$p_g = \frac{1}{P} \sum_{p=1}^P |y_g|^2 = \frac{1}{P} \sum_{p=1}^P \mathbf{u}_g^T \mathbf{x} \mathbf{x}^H \mathbf{u}_g^* = \mathbf{u}_g^T \mathbf{R} \mathbf{u}_g^*, \quad (9)$$

where

$$\mathbf{R} = \mathbb{E}\{\mathbf{x} \mathbf{x}^H\} = \mathbf{R}_F + \mathbf{R}_N + \sigma^2 \mathbf{I}_L \quad (10)$$

with $\mathbf{R}_F$ and $\mathbf{R}_N$ denoting the covariance matrices of FF and NF sources, respectively, and $\mathbf{I}_L$ is an $L \times L$ identity matrix.

To estimate the approximate position of FF sources, we first construct the weight vector:

$$\mathbf{u}_g = \mathbf{u}_f = [e^{j\frac{2\pi}{\lambda} P d \sin \varphi_g}, \dots, 1, \dots, e^{-j\frac{2\pi}{\lambda} P d \sin \varphi_g}]^T \quad (11)$$

and by leveraging the beamforming framework presented in [11], we deduce that when φ_g aligns with the direction of the true signal, the average power computed via Eq. (8) will exceed that corresponding to any other phase shift values.

An analogous principle applies to NF sources, where the weight vector is defined as:

$$\mathbf{u}_g = \mathbf{u}_n = [e^{j(\frac{2\pi}{\lambda} P d \sin \varphi_g + \pi \frac{P^2 d^2}{\lambda r} \cos^2 \varphi_g)}, \dots, 1, \dots, e^{j(-\frac{2\pi}{\lambda} P d \sin \varphi_g + \pi \frac{P^2 d^2}{\lambda r} \cos^2 \varphi_g)}]^T \quad (12)$$

By identifying the peak regions in the beamforming output, we can then obtain rough estimates of θ_k and range r_k for the k -th source signal.

In order to further improve the estimation accuracy of localization and finely control the phase of RIS reflection unit,

a refinement process near the initial estimates is employed here. Take the coarse DOA estimation as an example, for each DOA/angle θ_k estimated by the beamforming, a number of adjacent angles are selected on its both sides to form a refined angle set: $N(\theta_k) = \{\theta_k - \Delta, \dots, \theta_k + \Delta\}$, where Δ is the angular offset, which is usually set to 1° to 3° and can be set adaptively.

Define $\theta_1, \theta_2, \dots, \theta_G$ as a set of predetermined angles, which is used to calculate the corresponding weight vector for each angle in the set, so as to realize local fine beam scanning of the estimated direction [11], the same operations can be applied to the range and update the reflection phase of RIS to $\bar{\varphi}_1, \bar{\varphi}_2, \dots, \bar{\varphi}_G$ accordingly.

Subsequently, by defining the new weight vectors as $\bar{\mathbf{u}}_g = [e^{j\bar{\varphi}_{1,g}}, \dots, e^{j\bar{\varphi}_{L,g}}]$, where $\bar{\varphi}_{l,g}$ denotes the chosen phase at the l -th RIS element. By employing the well-known vectorization identity $\text{vec}(\mathbf{XYZ}) = (\mathbf{Z}^T \otimes \mathbf{X})\text{vec}(\mathbf{Y})$, Eq. (9) can then be rewritten as

$$\bar{p}_g = \text{vec}(\bar{\mathbf{u}}_g^T \mathbf{R} \bar{\mathbf{u}}_g^*) = (\bar{\mathbf{u}}_g^H \otimes \bar{\mathbf{u}}_g^T) \text{vec}(\mathbf{R}) \quad (13)$$

Further denote $\boldsymbol{\nu}_g = \bar{\mathbf{u}}_g^H \otimes \bar{\mathbf{u}}_g^T$ and $\boldsymbol{\rho} = \text{vec}(\mathbf{R})$, we have $\bar{p}_g = \boldsymbol{\nu}_g \boldsymbol{\rho}$. Taking G predetermined angles and/or ranges into consideration, the following overdetermined linear system can be constructed as

$$\boldsymbol{\nu} \boldsymbol{\rho} = \mathbf{p} \quad (14)$$

where $\boldsymbol{\nu} = (\boldsymbol{\nu}_1, \boldsymbol{\nu}_2, \dots, \boldsymbol{\nu}_G)^T$ is a $G \times L^2$ matrix and $\mathbf{p} = [p_1, p_2, \dots, p_G]^T$ is a $G \times 1$ vector.

Instead of solving Eq. (11) directly, the diagonal loading technique is applied here to avoid the possible ill-conditioned problem. Let κ denote the diagonal loading coefficient, then $\boldsymbol{\rho}$ can be estimated as

$$\hat{\boldsymbol{\rho}} = (\boldsymbol{\nu}^H \boldsymbol{\nu} + \kappa \mathbf{I}_{L^2})^{-1} \boldsymbol{\nu}^H \mathbf{p} \quad (15)$$

Finally, the SCM can be reconstructed by

$$\hat{\mathbf{R}} = \text{unvec}(\hat{\boldsymbol{\rho}}) \quad (16)$$

where $\text{unvec}(\mathbf{m})$ stands for the operation of inverse vectorization.

IV. MIXED SOURCE LOCALIZATION

A. NF Source Localization

With available SCM $\hat{\mathbf{R}}$, we exploit the symmetry of the RIS array to isolate NF and FF components. It should be noted that the covariance matrix $\mathbf{R}_F$ with respect to FF sources has a Hermitian Toeplitz structure, satisfying $\mathbf{R}_F = \mathbf{J} \mathbf{R}_F^T \mathbf{J}$, where $\mathbf{J}$ is an exchange matrix with ones on its antidiagonal and zeros elsewhere. In contrast, $\mathbf{R}_N$ corresponding to NF sources only retains a Hermitian form. This allows the spatial differencing scheme can be used directly to eliminate FF components, detailed as follows:

$$\mathbf{R}_D = \mathbf{R} - \mathbf{J} \mathbf{R}^T \mathbf{J} = \mathbf{R}_N - \mathbf{J} \mathbf{R}_N^T \mathbf{J}. \quad (17)$$

It can be observed that the FF components are significantly attenuated, and the resulting covariance matrix $\mathbf{R}_D$ mainly

contains the NF components. Performing eigenvalue decomposition (EVD) on $\mathbf{R}_D$ yields

$$\mathbf{R}_D = \mathbf{U}_S \mathbf{S}_s \mathbf{U}_S^H + \mathbf{U}_N \mathbf{S}_n \mathbf{U}_N^H \quad (18)$$

where $\mathbf{S}_s$ and $\mathbf{S}_n$ denote the diagonal matrix composed of signal eigenvalues and noise eigenvalues, respectively; $\mathbf{U}_S$ and $\mathbf{U}_N$ are the signal subspace matrix and noise subspace matrix, respectively.

With $\mathbf{U}_N$, the DOA and range of K_1 NF sources can be estimated by the following MUSIC spectral function:

$$P_{\text{MUSIC}}^{\text{NF}}(\theta, r) = \frac{1}{\mathbf{a}_{\text{NF}}^H(\theta, r) \mathbf{U}_N \mathbf{U}_N^H \mathbf{a}_{\text{NF}}(\theta, r)} \quad (19)$$

B. FF Source Localization

After estimating NF source parameters via matrix differencing technique, the remaining task is to estimate FF DOA parameters. Since the parameters of NF sources have been estimated, we apply the oblique projection operator to removes the NF components while preserving the FF components [12]. Let $\hat{\mathbf{A}}_N$ denote the reconstructed array manifold matrix of the NF sources, subsequently, the oblique projection matrix can be calculated by

$$\mathbf{Z}_N = \mathbf{I}_L - \hat{\mathbf{A}}_N (\hat{\mathbf{A}}_N^H \hat{\mathbf{A}}_N)^{-1} \hat{\mathbf{A}}_N^H \quad (20)$$

which directly yields that the covariance matrix only related to FF components can be written as

$$\hat{\mathbf{R}}_F = \mathbf{Z}_N \hat{\mathbf{R}} \mathbf{Z}_N^H. \quad (21)$$

By performing EVD on $\hat{\mathbf{R}}_F$ and subsequently conducting one-dimensional MUSIC spectral search, the DOA parameters of FF sources can be finally obtained. At this point, the NF and FF sources have been successfully classified, along with their respective parameter estimations.

V. NUMERICAL SIMULATIONS

This section evaluates the performance of the proposed method, and compare it with the TS-MUSIC and OP-MUSIC. A symmetric uniform linear array of 17 RIS elements with two FF sources at -20° and 30° , and two NF sources at $(50^\circ, \lambda)$ and $(-10^\circ, 2\lambda)$, and element spacing $d = 0.25\lambda$ is employed. Performance is quantified via root mean square error (RMSE) over 500 independent Monte Carlo trials.

In the first simulation, RMSEs of parameter estimations under different SNRs are assessed with 400 snapshots, and the simulation results are shown in Figs. 2, it can be observed that the RMSEs of both methods decrease as the increase of SNR. Moreover, the proposed one shows superior performance compared to the TS-MUSIC and OP-MUSIC, validating the effectiveness of the proposed method effectively.

In the second simulation, we further assess the performance of the proposed method under different number of snapshots, the simulation configurations are the same as the first experiment, except that the SNR is set at 10 dB, and the number of snapshots is varied from 200 to 1200. It can be seen from Figs. 3 that the RMSE of the proposed method

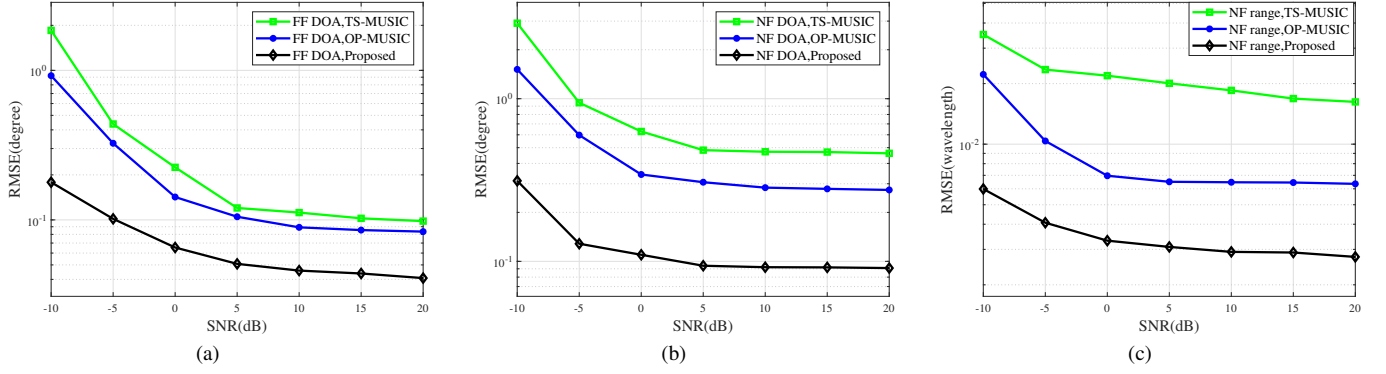


Fig. 2. RMSE curves versus SNR: (a) DOA estimation of FF sources; (b) DOA estimation of NF sources; (c) Range estimation of NF sources.

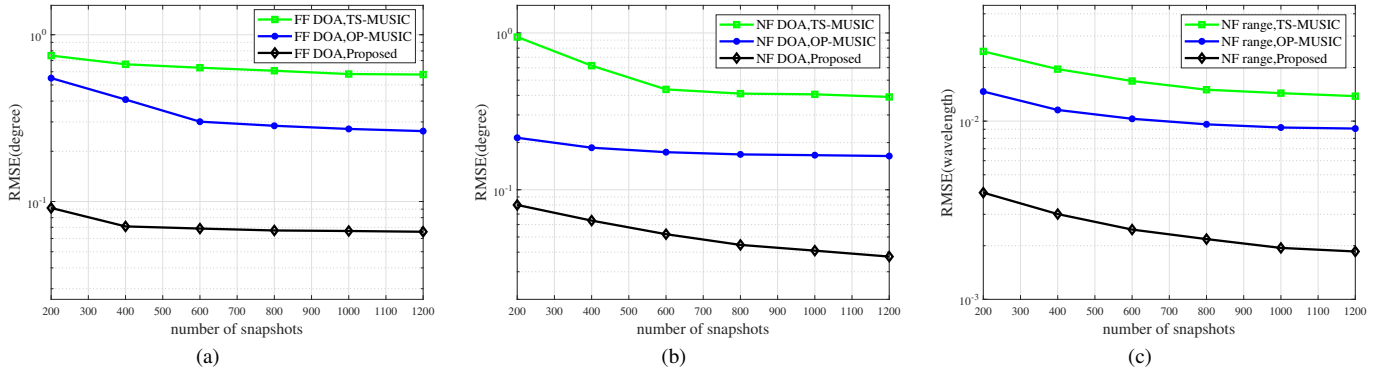


Fig. 3. RMSE curves versus the number of measurements: (a) DOA estimation of FF sources; (b) DOA estimation of NF sources; (c) Range estimation of NF sources.

decreases monotonically with the number of snapshots. More importantly, it provides an significantly improved DOA estimation performance in comparison with the TS-MUSIC and OP-MUSIC .

VI. CONCLUSION

This paper proposes a novel symmetric RIS-aided mixed-source localization method with a single receiving antenna. By exploiting the geometric symmetry of the RIS, the proposed method enables efficient utilization of RIS resources and reconstruction of high-dimensional SCM. Furthermore, the spatial differencing and oblique projection techniques are jointly employed to effectively isolate NF and FF source components, and subsequently achieve NF and FF source localization independently. Simulation results show that the proposed approach can improve the localization accuracy of mixed sources effectively.

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